The Besson–Courtois–Gallot Barycenter Method and Ruan's Octonionic Repair

A Machine-Checked Formalization Blueprint in Lean 4

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Contents

1 Spectral Linear Algebra and Volume Entropy 2

2 Hyperboloid Busemann Calculus and the Smooth Natural Map 4

3 General Rank-One Symmetric Spaces and Ruan's Octonionic Repair 6

4 Global Volume-Entropy Inequalities and Master Theorems 9

Chapter 1

Spectral Linear Algebra and Volume Entropy

Theorem 1.1 (Spectral trace and determinant formulas)**.** *Let $T : E \to_l [\mathbb{R}]E$ be a symmetric linear endomorphism on an n-dimensional real inner product space E. Then*

$$\operatorname{Tr}(T) = \sum_{i=1}^{n} \lambda_i(T), \qquad \det(I - T) = \prod_{i=1}^{n} (1 - \lambda_i(T)),$$

and if T is positive-semidefinite ($\langle v, Tv\rangle \geq 0$ for all $v \in E$), then $\lambda_i(T) \geq 0$ and $\det(T) \geq 0$.

Proof. Spectral diagonalization in an orthonormal eigenbasis together with weighted AM–GM on $\operatorname{Fin} n$. □

Theorem 1.2 (Sharp AM–GM trace-determinant inequality)**.** *For any symmetric positive-semidefinite endomorphism $T : E \to_l [\mathbb{R}]E$ on an n-dimensional real inner product space ($n \geq 1$),*

$$\det(T) \leq \left(\frac{\operatorname{Tr}(T)}{n}\right)^n.$$

When $\operatorname{Tr}(H_1) \leq 1$ and $\operatorname{Tr}(H_2) \leq 1$, any $c \geq 0$ satisfies $c^n \sqrt{\det H_1}\sqrt{\det H_2} \leq (c/n)^n$.

Proof. Applies product-power AM–GM to the non-negative eigenvalues $\lambda_i(T)$. □

Theorem 1.3 (Adjoint-Gram determinant bound)**.** *For any linear operator $A : E \to_l [\mathbb{R}]E$ with $\|Av\| \leq c\|v\|$, the Gram operator A^*A satisfies $\det(A^*A) = (\det A)^2$, $\operatorname{Tr}(A^*A) = \sum_i \|Ae_i\|^2 \leq nc^2$, and $|\det A| \leq c^n$. Moreover, any symmetric positive-definite H ($\det H > 0$) admits a symmetric inverse square root $D = H^{-1/2}$ satisfying $D^2H = \mathrm{id}_E$ and $\det D = (\sqrt{\det H})^{-1}$.*

Proof. Spectral calculus on the orthonormal eigenbasis of A^*A and H. □

Theorem 1.4 (Souto's bilinear Cauchy–Schwarz determinant bound)**.** *Let $L, H_1, H_2 : E \to_l [\mathbb{R}]E$ on an n-dimensional real inner product space ($n \geq 1$) with H_1, H_2 symmetric positive-semidefinite and $c \geq 0$. If $|\langle Lu, w\rangle| \leq c\sqrt{\langle u, H_1 u\rangle}\sqrt{\langle w, H_2 w\rangle}$ for all $u, w \in E$, then*

$$|\det L| \leq c^n \sqrt{\det H_1}\sqrt{\det H_2}.$$

If furthermore $\operatorname{Tr}(H_1) \leq 1$ and $\operatorname{Tr}(H_2) \leq 1$, then $|\det L| \leq (c/n)^n$.

Proof. Normalizes L on the right and left by $H_1^{-1/2}$ and $H_2^{-1/2}$ when $\det H_i > 0$, reducing to $\|A\| \le c$, and handles the degenerate case $\det H_i = 0$ via kernel vectors. □

Definition 1.5 (Exponential distance weights and critical exponent)**.** For a metric-measure space (X, d, μ), scale $s \in \mathbb{R}$, and basepoint $x \in X$, the exponential weight is $w_s(x, z) = e^{-sd(x,z)}$, the critical exponent (volume entropy) is

$$h(X, \mu, x_0) = \inf \left\{ s \in \mathbb{R} \;\middle|\; \int_X e^{-sd(x_0,z)}\, d\mu(z) < \infty \right\},$$

and the exponentially weighted measure is $\mu_x^s = e^{-sd(x,\cdot)}\mu$.

Theorem 1.6 (Basepoint invariance of volume entropy)**.** *For any $s \ge 0$ and $x, y, z \in X$, the triangle inequality gives $e^{-sd(x,z)} \le e^{sd(x,y)}e^{-sd(y,z)}$. Consequently, $z \mapsto e^{-sd(x,z)}$ is μ-integrable if and only if $z \mapsto e^{-sd(y,z)}$ is μ-integrable, and $h(X, \mu, x) = h(X, \mu, y)$ is independent of the basepoint.*

Proof. Mutual domination of the integrands by the constant factor $e^{sd(x,y)}$. □

Chapter 2

Hyperboloid Busemann Calculus and the Smooth Natural Map

Definition 2.1 (Hyperboloid Busemann function and differential)**.** For a boundary direction $\xi \in S(E)$ and $y = (t, x) \in \mathbf{H}^n = \text{Hyperboloid}(E)$, the Busemann function is $B_\xi(y) = -\log(t - \langle x, \xi\rangle)$ with exponential $e^{B_\xi(y)} = (t - \langle x, \xi\rangle)^{-1} > 0$ and fiberwise differential $dB_\xi(y)$.

Theorem 2.2 (Exact Busemann Hessian identity on the hyperboloid)**.** *On the Lorentzian hyperboloid model $\mathbf{H}^n = \text{Hyperboloid}(E)$ equipped with its Levi-Civita connection, B_ξ is C^∞ and its Riemannian Hessian satisfies the exact identity*

$$(\operatorname{Hess} B_\xi)_y(X, Y) = g_y(X, Y) - dB_\xi(y)(X)\, dB_\xi(y)(Y).$$

In particular, $(dB_\xi(y)(X))^2 \le g_y(X, X)$ and $(\operatorname{Hess} B_\xi)_y(X, X) \ge 0$.

Proof. Bridges with `busemann_hessian_eq` on the hyperboloid model. □

Definition 2.3 (Busemann covariance and cross-operators)**.** For vectors $u, u_1, u_2 \in E$, define the rank-one operators $(u \otimes u^*)(v) = \langle u, v\rangle u$ and $(u_1 \otimes u_2^*)(v) = \langle u_1, v\rangle u_2$. Given finite weights w_a and vector families $u_a, u_{1,a}, u_{2,a} \in E$, define the Busemann covariance operator and scaled cross-operator:

$$H(w, u) = \sum_{a \in s} w_a (u_a \otimes u_a^*), \qquad L_c(w, u_1, u_2) = c \sum_{a \in s} w_a (u_{1,a} \otimes u_{2,a}^*).$$

Theorem 2.4 (Busemann cross-operator determinant bound)**.** *For non-negative weights $w_a \ge 0$ with $\sum_{a \in s} w_a \le 1$ and unit-bounded vectors $\|u_{1,a}\| \le 1$, $\|u_{2,a}\| \le 1$, the covariance operators $H(w, u_i)$ are symmetric positive-semidefinite with $\operatorname{Tr}(H(w, u_i)) \le 1$, the cross-operator satisfies $|\langle L_c(w, u_1, u_2)v, z\rangle| \le c\sqrt{\langle v, H_1 v\rangle}\sqrt{\langle z, H_2 z\rangle}$, and*

$$|\det L_c(w, u_1, u_2)| \le \left(\frac{c}{n}\right)^n.$$

Proof. Discrete Cauchy–Schwarz on $\sum_a (\sqrt{w_a}\langle u_{1,a}, v\rangle)(\sqrt{w_a}\langle u_{2,a}, z\rangle)$ combined with Theorem 1.4. □

Definition 2.5 (Lorentzian Busemann barycenter functional and explicit minimizer)**.** For weights $w : \iota \to \mathbb{R}$ and boundary directions $\xi : \iota \to S(E)$, the Busemann barycenter functional on $\text{Hyperboloid}(E)$ is $\mathcal{B}_w(y) = \sum_{i \in \iota} w_i e^{B_{\xi_i}(y)} = -\langle y, Z(w)\rangle_{\mathcal{L}}$, where $Z(w) = (\sum_i w_i, \sum_i w_i \xi_i) \in$

$\mathbb{R} \times E$ is the resultant Lorentz vector. When $\|\sum_i w_i \xi_i\| < \sum_i w_i$, the future-timelike normalization hyperboloidBarycenter$(w, \xi) = Z(w)/\|Z(w)\|_{\mathcal{L}} \in$ Hyperboloid(E) defines the explicit barycenter, with normalized probability weights $p_i(w) = w_i e^{B_{\xi_i}(p)}/\mathcal{B}_w(p)$.

Theorem 2.6 (Closed-form uniqueness, C^∞ regularity, and Γ-equivariance of the barycenter)**.** *For any weight vector w with $\|\sum_i w_i \xi_i\| < \sum_i w_i$:*

1. *$\mathcal{B}_w(y) = \|Z(w)\|_{\mathcal{L}} \cosh d_{\mathbf{H}}(y, \mathrm{bar}(w))$, so $p =$ hyperboloidBarycenter(w, ξ) is the unique global minimizer of $\mathcal{B}_w$ on $\mathbf{H}^n$ and satisfies $\nabla \mathcal{B}_w(p) = 0$ and $d\mathcal{B}_w(p) = 0$.*
2. *The barycenter map $w \mapsto$ hyperboloidBarycenter(w, ξ) is C^∞ on the open future-timelike cone, is $\mathrm{Isom}(\mathbf{H}^n)$-equivariant under boundary cocycle rescaling, and descends to a continuous map on quotient manifolds $X/\Gamma_1 \to Y/\Gamma_2$.*

Proof. Hyperbolic cosine law in Minkowski space together with the implicit/explicit differentiation of $Z(w)/\|Z(w)\|_{\mathcal{L}}$. □

Definition 2.7 (Metric gauge isometry and Riemannian barycenter differential)**.** At $p = (t, x) \in$ Hyperboloid(E), the metric gauge operator $\Phi_p = I - (1 - t^{-1})P_x : E \simeq_l [\mathbb{R}]E$ with inverse $\Phi_p^{-1} = I + (t - 1)P_x$ is an exact linear isometry between the Riemannian tangent metric g_p and the Euclidean inner product on E. The orthonormalized Euclidean Busemann gradient is $u_2(\xi, p) = \Phi_p(\nabla B_\xi(p))$ ($\|u_2(\xi, p)\| = 1$), and the Riemannian differential of the natural map at scale s is $D^{\mathrm{Riem}} F_s(p)$.

Theorem 2.8 (Implicit differentiation and hyperbolic Jacobian bound)**.** *In the orthonormal gauge Φ_p, the Fréchet derivative of the hyperboloid barycenter map satisfies*

$$D^{\mathrm{Riem}} F_s(p) = L_s\big(p(w), u_1, u_2(\cdot, p)\big),$$

and for any $s \geq 0$ and unit-bounded source gradients $\|u_{1,i}\| \leq 1$,

$$\left|\det D^{\mathrm{Riem}} F_s(p)\right| \leq \left(\frac{s}{n}\right)^n.$$

Proof. Differentiates $Z(w)/\|Z(w)\|_{\mathcal{L}}$ in spatial chart coordinates, pulls back by Φ_p, and applies Theorem 2.4. □

Theorem 2.9 (Spectral AM–GM equality and hyperbolic rigidity)**.** *If $z_i \geq 0$ on $\mathrm{Fin}\, n$ satisfy $\sum_{i=1}^n z_i \leq 1$ and $\prod_{i=1}^n z_i = (1/n)^n$, then $z_i = 1/n$ for all i. Consequently, if $|\det D^{\mathrm{Riem}} F_s(p)| = (s/n)^n$ with $s > 0$, then:*

1. *Both source and target Busemann covariance tensors are isotropic: $H_1 = H_2 = \frac{1}{n}\,\mathrm{id}_E$.*
2. *$(D^{\mathrm{Riem}} F_s(p))^* D^{\mathrm{Riem}} F_s(p) = (s/n)^2\,\mathrm{id}_E$, so $D^{\mathrm{Riem}} F_s(p)$ is a scaled linear isometry with scale factor s/n (and an exact Riemannian linear isometry when $s = n$).*

Proof. Strict concavity of $\log t \leq t - 1$ forces all eigenvalues of H_1, H_2 and A^*A to be equal, and L^1 saturation of $\int_X (f - g)\, d\mu_X = 0$ with $f \leq g$ upgrades global volume equality to μ_X-a.e. pointwise equality. □

Chapter 3

General Rank-One Symmetric Spaces and Ruan's Octonionic Repair

Definition 3.1 (Rank-one symmetric space invariants and $\mathbb{K}$-line projections)**.** For a rank-one symmetric space $\widetilde{M} = \mathbb{K}\mathbf{H}^m$ ($\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$, $d = \dim_{\mathbb{R}} \mathbb{K} \in \{1, 2, 4, 8\}$, real dimension $n = dm$, curvature $K \in [-4, -1]$), the volume entropy is $h(\mathbb{K}\mathbf{H}^m) = n + d - 2$. For a unit Busemann gradient $u \in E$ and orthogonal projection π_u onto the d-dimensional $\mathbb{K}$-line $\mathbb{K}u$, the single-direction Busemann Hessian is $W(u, \pi_u) = I - (u \otimes u^*) - (\pi_u - u \otimes u^*) = I - 2(u \otimes u^*) + \pi_u$, and the integrated Hessian operator is

$$W_y = I - 2H + \widehat{H} = (I - H) + (\widehat{H} - H), \qquad \widehat{H} = \sum_{a \in s} w_a \pi_{u_a}.$$

Theorem 3.2 (Riccati square identity and implicit differentiation reduction)**.** *Every $\mathbb{K}$-line Busemann Hessian satisfies the Riccati equation $W(u, \pi_u)^2 = -R(\cdot, u)u = 4(\pi_u - u \otimes u^*) + (I - \pi_u)$ and $\operatorname{Tr}(W(u, \pi_u)) = n + d - 2$. Moreover, $\widehat{H} - H \geq 0$ with $\operatorname{Tr}(\widehat{H}) = d$, and whenever $W_y \circ D_F = sH_y$,*

$$|\det D_F| \leq \left(\frac{s}{\sqrt{n}}\right)^n \frac{\sqrt{\det H}}{\det(I - 2H + \widehat{H})}.$$

Consequently, the rank-one determinant inequality $\frac{\sqrt{\det H}}{\det(I - 2H + \widehat{H})} \leq \left(\frac{\sqrt{n}}{h_0}\right)^n$ implies $|\det D_F| \leq (s/h_0)^n$, and the quantitative eigenvalue-defect version implies isotropy $H = \frac{1}{n} \operatorname{id}_E$ upon equality.

Proof. Orthogonal decomposition into $\mathbb{R}u \oplus (\mathbb{K}u \ominus \mathbb{R}u) \oplus (\mathbb{K}u)^\perp$ with eigenvalues $(0, 2, 1)$, combined with Hölder/Cauchy–Schwarz on H_y. □

Theorem 3.3 (Minkowski–Ky Fan log-concavity of the determinant (BCG Lemma B.2))**.** *For any symmetric positive-semidefinite operators $A, B : E \to_l [\mathbb{R}]E$ on an n-dimensional real inner product space and weights $p, q \geq 0$ with $p + q = 1$,*

$$\det(p \bullet A + q \bullet B) \geq (\det A)^p (\det B)^q.$$

By induction, if $A_0, \ldots, A_{m-1}$ are symmetric positive-semidefinite with $\det A_r \geq d_0 \geq 0$, then $\det\left(\frac{1}{m} \sum_{r=0}^{m-1} A_r\right) \geq d_0$.

Proof. Simultaneous diagonalization via $A^{-1/2}$ ($DAD = \mathrm{id}_E$), reducing to $\prod_{i=1}^n (p + q\mu_i) \geq \prod_{i=1}^n \mu_i^q$ for the eigenvalues $\mu_i \geq 0$ of DBD. □

Definition 3.4 (Global orthogonal $\mathbb{K}$-structure for associative division algebras)**.** For the associative division algebras $\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$ ($d \in \{1, 2, 4\}$), an orthogonal $\mathbb{K}$-structure on $E \cong T_x \mathbb{K}\mathbf{H}^m$ is a family $J_0 = \mathrm{id}_E, J_1, \ldots, J_{d-1} \in \mathrm{O}(E)$ with $\langle J_r u, J_s u \rangle = 0$ for $r \neq s$, inducing the $\mathbb{K}$-line projection $\pi_u = \sum_{r=0}^{d-1} (J_r u) \otimes (J_r u)^*$.

Theorem 3.5 (Associative rank-one determinant inequality and BCG Jacobian bound)**.** *For an associative* `IsOrthogonalKStructure d J` *($d \in \{2, 4\}$):*

1. *Property* (P2) *holds globally:* $\widehat{H} - H = \sum_{r \neq 0} J_r H J_r^*$*, and hence* $\det((\widehat{H} - H)/(d-1)) \geq \det H$ *(BCG Lemma B.3).*

2. *Combining the log-concavity splitting* $W_y = (n + d - 2)\big(p \frac{I - H}{n-1} + q \frac{\widehat{H} - H}{d-1}\big)$ *($p = \frac{n-1}{n+d-2}$, $q = \frac{d-1}{n+d-2}$) with BCG Lemma B.4 yields*

$$\frac{\sqrt{\det H}}{\det(I - 2H + \widehat{H})} \leq \left(\frac{\sqrt{n}}{n + d - 2}\right)^n \quad \text{and} \quad |\det D_F| \leq \left(\frac{s}{n + d - 2}\right)^n.$$

Proof. Applies Theorem 3.3 to the $d-1$ orthogonal conjugates $J_r H J_r^*$ (each having determinant $\det H$) and combines with the spectral AM–GM bound for $(I - H)/(n-1)$. □

Definition 3.6 (Local Cayley-line orthogonal structure on $\mathbb{O}\mathbf{H}^2$)**.** Although non-associativity of the octonions $\mathbb{O}$ prevents a global `IsOrthogonalKStructure 8 J` on $T_x \mathbb{O}\mathbf{H}^2 \cong \mathbb{R}^{16}$, on each individual 8-dimensional Cayley line $V = \mathcal{C}ay(v_i) \cong \mathbb{R}^8$ there exist line-dependent orthogonal endomorphisms $J_{0,v_i} = \mathrm{id}_V, J_{1,v_i}, \ldots, J_{7,v_i} \in \mathrm{O}(V)$ generated by octonionic right multiplication (Ruan Lemma 3.2 (P2′)).

Theorem 3.7 (Ruan's Cayley block trace identity and 8×8 block AM–GM bounds)**.** *On the 16-dimensional Cayley hyperbolic plane* $\mathbb{O}\mathbf{H}^2$ *($n = 16$, $d = 8$, $h_0 = 22$, critical ratio $\sqrt{16}/22 = 2/11$):*

1. *By Ruan Lemma 3.2* (P1′)–(P2′)*, on any 8-dimensional Cayley line V where $\widehat{H}|_V = \lambda \mathrm{id}_V$ and $\lambda \|x\|^2 = \sum_{r=0}^7 \langle J_r^* x, A(J_r^* x) \rangle$, polarization and cyclic trace invariance yield* $\mathrm{Tr}(A) = \lambda$ *(and $\mathrm{Tr}(B) = \mu$, $\lambda + \mu = 1$).*

2. *Rescaling $A' = \lambda^{-1} A$ on V ($\dim_{\mathbb{R}} V = 8$) and applying* `det_id_sub_div_ge_det` *proves Ruan's Eq. (28):*

$$\det(\lambda I_8 - A) \geq 7^8 \det A \qquad \text{and} \qquad \det(\mu I_8 - B) \geq 7^8 \det B.$$

Proof. Polarizes $\sum_{r=0}^7 J_r A J_r^* = \lambda \mathrm{id}_V$, takes traces using $\mathrm{Tr}(J_r A J_r^*) = \mathrm{Tr}(A)$, and applies the $s = 1$ spectral AM–GM theorem in dimension $d = 8$. □

Theorem 3.8 (Ruan's Case 1 Schur monotonicity and Case 2 reverse AM–GM interpolation)**.** *In Ruan's two-case analysis of the 8×8 block decomposition* $H = \begin{pmatrix} A & C \\ C^* & B \end{pmatrix}$*:*

1. ***Case 1*** *(*$2\lambda_1 \le \lambda$***,*** $2\mu_1 \le \mu$***,*** $\alpha = 0$*):* *For* $0 \le w_i \le h_{2,i} \le h_{1,i}$, $\prod_i (h_{2,i} - w_i) \prod_i h_{1,i} \le \prod_i (h_{1,i} - w_i) \prod_i h_{2,i}$ *(Ruan Lemma 3.5). Combined with* $\det(\lambda I_8 - A) \ge 7^8 \det A$ *and* $\det(\mu I_8 - B) \ge 7^8 \det B$*, Schur complement factorization yields* $\det(\widehat{H} - H) \ge 7^{16} \det H$*.*

2. ***Case 2*** *(*$2\mu_1 > \mu$ ***or*** $2\lambda_1 > \lambda$***,*** $\alpha \in [1 - \sqrt{2/3}, 0.266]$*):* *Every non-negative* $x : \operatorname{Fin} 8 \to \mathbb{R}$ *satisfies the 8-dimensional reverse AM–GM inequality* $7^7 \prod_{i=0}^{7} x_i \le (\sum_{i=0}^{7} x_i - x_0)^7 x_0$ *(Ruan Lemma 3.6), and* $t = 1 - (1 - \alpha)^2$ *satisfies* $t^2(1 - t) \ge 7/64$*, yielding the cubic endpoint bound* $\frac{7}{64}\mu(\mu - \mu_1)\mu_1 \le (\mu - t\mu)(t\mu)^2$*.*

Proof. Pointwise algebraic monotonicity for Case 1 and 7-variable weighted AM–GM on $\operatorname{Fin} 8 \setminus \{0\}$ for Case 2. □

Theorem 3.9 (Ruan's perturbed log-concavity trick and Proposition 3.1 on $\mathbb{O}\mathbf{H}^2$)**.** *For any off-diagonal perturbation operator* $U_\alpha : E \to_l [\mathbb{R}]E$ *such that* $I - H - U_\alpha \ge 0$ *and* $\widehat{H} - H + U_\alpha \ge 0$*, the Hessian operator splits as* $I - 2H + \widehat{H} = (I - H - U_\alpha) + (\widehat{H} - H + U_\alpha)$ *and satisfies Ruan's perturbed log-concavity bound (Eq. (22)):*

$$\det(I - 2H + \widehat{H}) \ge 22^{16} \det\left(\frac{I - H - U_\alpha}{15}\right)^{15/22} \det\left(\frac{\widehat{H} - H + U_\alpha}{7}\right)^{7/22}.$$

Combined with Ruan Lemma 3.4 (Eqs. (21) and (23)), this proves Proposition 3.1 and Corollary 3.8 on $\mathbb{O}\mathbf{H}^2$*:*

$$\frac{\sqrt{\det H}}{\det(I - 2H + \widehat{H})} \le \left(\frac{\sqrt{16}}{22}\right)^{16} = \left(\frac{2}{11}\right)^{16} \quad \text{and} \quad |\det D_F| \le \left(\frac{s}{22}\right)^{16}.$$

Proof. Applies Theorem 3.3 with $p = 15/22, q = 7/22$ to $(I - H - U_\alpha)/15$ and $(\widehat{H} - H + U_\alpha)/7$, then combines exponents $\frac{4}{15} \cdot \frac{15}{22} + \frac{7}{22} = \frac{11}{22} = \frac{1}{2}$. □

Chapter 4

Global Volume-Entropy Inequalities and Master Theorems

Theorem 4.1 (Global BCG volume-entropy integration and scale limit $s \searrow h(X)$)**.** *Let (X, μ_X) be a finite-measure space. If at every scale $s > h_X$, the Jacobian $J_s(x) = |\det d_x F_s|$ is integrable, bounded μ_X-almost everywhere by $(s/h_Y)^n$, and satisfies the degree-area inequality $|\deg(f)| \cdot \mathrm{Vol}(Y) \le \int_X J_s(x)\, d\mu_X(x)$, then*

$$|\deg(f)| \cdot h_Y^n \cdot \mathrm{Vol}(Y) \le s^n \cdot \mathrm{Vol}(X) \quad (\forall s > h_X) \qquad \Longrightarrow \qquad |\deg(f)| \cdot h_Y^n \cdot \mathrm{Vol}(Y) \le h_X^n \cdot \mathrm{Vol}(X).$$

Proof. Monotonicity of Bochner integration on (X, μ_X) followed by right-continuity of $s \mapsto s^n \mathrm{Vol}(X)$ along the punctured neighborhood filter $\mathcal{N}_{>}(h_X)$. □

Theorem 4.2 (Master Smooth Real Hyperbolic BCG Volume-Entropy and Rigidity Theorems)**.** *For the explicit smooth natural map $F_s(x) = \mathrm{hyperboloidBarycenter}(w(s,x), \xi)$ on an n-dimensional hyperbolic target ($n \ge 1$), integrating $|\det D^{\mathrm{Riem}} F_s(x)| \le (s/n)^n$ and passing to $s \searrow c_X$ yields*

$$|\deg(f)| \cdot n^n \cdot \mathrm{Vol}(Y) \le c_X^n \cdot \mathrm{Vol}(X).$$

Moreover, if equality holds at scale $s > 0$, then for μ_X-almost every $x \in X$, $|\det D^{\mathrm{Riem}} F_s(x)| = (s/n)^n$, both Busemann covariance tensors are isotropic ($\frac{1}{n}\,\mathrm{id}_E$), and $D^{\mathrm{Riem}} F_s(x)$ is a (s/n)-scaled linear isometry.

Proof. Synthesizes Theorem 2.8, Theorem 2.9, and Theorem 4.1. □

Theorem 4.3 (Master General Rank-One and Cayley Hyperbolic Plane BCG Theorems)**.** *For any rank-one locally symmetric target (Y, g_0) modeled on $\mathbb{K}\mathbf{H}^m$ ($\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$, $n = \dim_{\mathbb{R}} Y$, $d = \dim_{\mathbb{R}} \mathbb{K}$, $h_0 = n + d - 2$), and in particular for the 16-dimensional Cayley hyperbolic plane $\mathbb{O}\mathbf{H}^2$ ($n = 16$, $d = 8$, $h_0 = 22$) via Ruan's perturbed log-concavity repair, the global Besson–Courtois–Gallot volume-entropy inequality holds:*

$$|\deg(f)| \cdot h_0^n \cdot \mathrm{Vol}(Y) \le c_X^n \cdot \mathrm{Vol}(X), \qquad |\deg(f)| \cdot 22^{16} \cdot \mathrm{Vol}(Y) \le c_X^{16} \cdot \mathrm{Vol}(X).$$

Proof. Combines Theorem 3.2, Theorem 3.5, and Theorem 3.9 with Theorem 4.1. □